EXPERIMENT No 4:

Uncontrolled Robot (Forward Robot car Experiment)

Student Guide

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Objective:
In this experiment, you will create a simple robot car that moves forward using 4 motors, 4 wheels, a single switch, and a battery. This activity will help you understand the basic concepts of motor control and wiring in a basic robotic system.

Materials Needed:

- 4 DC motors
- 4 wheels (one attached to each motor)
- 1 switch (SPST - Single Pole Single Throw)
- 1 battery (sufficient to power all motors)
- Wires for connections
- Chassis or base for attaching motors and wheels
- Tools (screwdrivers, wire cutters)

Theory:

- DC Motors are used to convert electrical energy into mechanical motion. When connected to a power source (battery), they spin, causing the attached wheels to rotate.

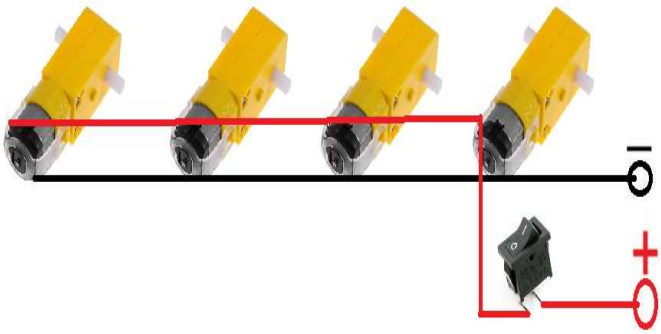
- SPST Switch is a simple switch that allows you to turn the motor circuit on or off. In this experiment, the switch will control whether the motors receive power or not, thus controlling the movement of the robot.

Steps for Building the Forward-Only Robot Car:

1. Assemble the Chassis:
   - Attach the 4 DC motors to the chassis or base. Each motor should be connected to a wheel so that all four wheels can rotate when the motor is powered.
2. Wiring the Motors:
   - Connect the positive and negative terminals of each motor together in parallel.
   - This ensures that when the battery is connected, all four motors will receive power and spin in the same direction, driving the robot forward.
3. Connect the Switch:
   - Connect one terminal of the switch to the positive terminal of the battery.
   - Connect the other terminal of the switch to the positive wires of the motors.

- Connect the negative terminal of the battery directly to the negative wires of the motors.

Circuit Diagram:



Test the Circuit:

- Once the switch is connected to the motors and battery, toggle the switch to the "on" position. The motors should start spinning, and the robot will move forward.

4. Mount the Components:
   - Secure the switch, battery, and wiring neatly onto the chassis to ensure all components are stable.
5. Run the Robot:
   - Place the robot on a flat surface and toggle the switch on. The car should move forward continuously as long as the switch is in the "on" position.

Observations:

- What happens when you turn the switch on?
- Does the robot move smoothly forward, or are there any issues with the movement?
- Can you identify which parts of the wiring control the flow of power to the motors?

Conclusion:

In this experiment, you created a simple robot that moves forward using basic motor control. The motors were wired in parallel to ensure equal power distribution, and the switch allowed you to control the robot's movement. This experiment demonstrates the fundamentals of DC motor control and wiring for simple robotic applications.

Challenge:

- Can you modify the design to include more functionality, like controlling the robot's direction?
- What happens if you change the number of motors or their arrangement?